An Empirical Note on LLL Degeneracy of the Boneh–Venkatesan Lattice for secp256k1

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Abstract

We document two distinct LLL-convergence failure modes for the Boneh–Venkatesan Hidden-Number Problem (HNP) lattice on deployed ECDSA curves. **Failure mode A**: at 256-bit, LLL fails for secp256k1 while succeeding for P-256 and brainpoolP256r1 at identical parameters (10 s timeout vs. 0.7 s convergence) — a curve-specific arithmetic phenomenon. **Failure mode B**: at $\geq$ 384 bits, LLL fails uniformly across P-384, brainpoolP384r1, and P-521 — consistent with the iteration cap being undersized relative to the lattice's expected convergence time at large bit-lengths. The initial hypothesis (near-power-of-2 group order is the cause) is refuted by the smaller Koblitz curves `secp192k1` and `secp224k1` succeeding at LLL despite both having near-power-of-2 group orders. We provide the empirical table, a sketched Gram–Schmidt analysis identifying μ-oscillation as the likely Failure-A mechanism, and concrete follow-up experiments to disambiguate the remaining hypotheses.

1 Setting and notation

The Boneh–Venkatesan lattice attack on biased-nonce ECDSA [1, 2] recovers the secret key d from m signatures (r_i, s_i, z_i) in which the nonce k_i is known to satisfy $0 \leq k_i < 2^{k_{\text{bits}}}$ for some $k_{\text{bits}} < n_{\text{bits}}$. With $a_i = s_i^{-1} r_i \bmod n$ and $t_i = s_i^{-1} z_i \bmod n$, each signature contributes the constraint

$$d \cdot a_i + t_i \equiv k_i \pmod{n}.$$

The standard $(m+2)$-dimensional integer lattice basis (rows)

$$B = \begin{pmatrix} n^2 \cdot I_m & 0_{m\times 2} & \\ n \cdot \mathbf{a}^\top & 2^{k_{\text{bits}}} & 0 \\ n \cdot \mathbf{t}^\top & 0 & n \cdot 2^{k_{\text{bits}}} \end{pmatrix}$$

is fed to LLL with Lovász parameter $\delta = 0.75$; the short vector after reduction contains $-d \cdot 2^{k_{\text{bits}}}$ in the $(m+1)$-st coordinate (modulo sign).

2 The phenomenon

We ran identical attack parameters head-to-head on two 256-bit deployed curves: NIST P-256 [5] and secp256k1 [6]. Both have prime group order n of bit-length 256.

2.1 Empirical measurements

Three independent deterministic seed pairs $(d_{\text{seed}}, k_{\text{seed}})$ were tested. The exact LLL implementation, δ, iteration cap, and Gram–Schmidt order were identical for both curves. Signatures were generated by a deterministic ECDSA-with-injected-bias signer.

Additional probes (BKZ-10, larger m, varying k_{bits}) reproduce the same qualitative pattern: secp256k1 lattices fail where P-256 lattices succeed.

2.2 Reproducibility

The probe is committed at

`tests/lll_degeneracy_probe.rs`

of the companion repository and exits in approximately 53 seconds on commodity hardware. The HNP attack itself lives in `src/cryptanalysis/hnp_ecdsa.rs`; we use it unmodified.

$(d_{\text{seed}}, k_{\text{seed}})$	P-256 outcome	P-256 time	secp256k1 outcome
`(0xC0FFEE, 0xC0FFEE)`	✓recovered	690 ms	iteration cap @ 9993 ms
`(0xDEADBEEF, 0xBADCAFE)`	✓recovered	697 ms	iteration cap @ 9845 ms
`(0x12345678, 0x9ABCDEF0)`	✓recovered	646 ms	iteration cap @ 10119 ms

Table 1: LLL behaviour: $k_{\text{bits}} = 192$, $m = 8$, $\delta = 0.75$. P-256: 3/3 recoveries. secp256k1: 0/3 recoveries.

3 Initial hypothesis and its refutation

The most salient arithmetic difference between secp256k1 and P-256 is the structure of their group orders. P-256's n is

$$n_{\text{P-256}} = \texttt{0xFFFFFFFF00000000FFFFFFFFFFFFFFFFBCE6FAADA7179E84F3B9CAC2FC632551},$$

while secp256k1's n is

$$n_{\text{secp256k1}} = \texttt{0xFFFFFFFF...FEBAAEDCE6AF48A03BBFD25E8CD0364141}$$

which can be written as $n = 2^{256} - t_{\text{secp}}$ for $t_{\text{secp}} \approx 2^{129}$. secp256k1's group order has a near-power-of-2 form with the deficit concentrated in the bottom 129 bits.

Our initial working hypothesis: **the Boneh–Venkatesan basis matrix, which has entries n^2 and $n \cdot a_i$, produces a Gram–Schmidt orthogonalisation with ratios that concentrate in LLL-pathological values when n has near-power-of-2 form, while generic n avoids this.** Three testable predictions followed:

1. Other near-power-of-2-n Koblitz curves should exhibit the same LLL-degeneracy: `secp192k1`, `secp224k1`.
2. Generic-n curves (P-256, brainpoolP256r1) should consistently converge.
3. BKZ-β with $\beta \geq 20$ may patch the degeneracy.

3.1 Cross-Koblitz refutation

We extended the head-to-head probe to `secp192k1` (Koblitz, 192-bit, $n \approx 2^{192}$) and `secp224k1` (Koblitz, 224-bit, $n \approx 2^{224}$). Both are *also* near-power-of-2 group orders. If the initial hypothesis were correct, both should fail LLL similarly to secp256k1.

Curve	n bit-length	LLL outcome (2 seeds)	Time per probe
`secp192k1`	192	2/2 RECOVERED	∼320 ms
`secp224k1`	224	2/2 RECOVERED	∼630 ms
`secp256k1`	256	0/2 (iteration cap)	∼10000 ms
`P-256`	256	2/2 RECOVERED	∼700 ms

Table 2: Cross-Koblitz LLL probe. Parameters scaled to each curve's bit-length: $k_{\text{bits}} = 3n_{\text{bits}}/4$, $m = 8$. The two smaller Koblitz curves (secp192k1, secp224k1) – both of which have near-power-of-2 group orders – succeed at LLL where secp256k1 alone fails.

The initial hypothesis is refuted. The LLL-degeneracy is not a general feature of near-power-of-2 group orders, nor of Koblitz curves more broadly. It is specific to secp256k1.

4 Extended bit-length sweep: two distinct degeneracies

A wider sweep including P-384, P-521, brainpoolP256r1, and brainpoolP384r1 reveals that there are *two* distinct degeneracies, not one:

Curve	n bits	LLL outcome	Time per probe	Notes
`secp192k1`	192	RECOVERED	∼389 ms	Koblitz, $j = 0$
`secp224k1`	224	RECOVERED	∼617 ms	Koblitz, $j = 0$
`secp256k1`	256	iteration cap	∼10000 ms	Koblitz, $j = 0$ **(outlier)**
`P-256`	256	RECOVERED	∼710 ms	NIST, generic j
`brainpoolP256r1`	256	RECOVERED	∼714 ms	generic j
`P-384`	384	iteration cap	∼20392 ms	NIST, generic j **(uniform fail)**
`brainpoolP384r1`	384	iteration cap	∼20131 ms	generic j **(uniform fail)**
`P-521`	521	iteration cap	∼20662 ms	NIST, generic j **(uniform fail)**

Table 3: Extended LLL-convergence sweep, $k_{\text{bits}} = 3n_{\text{bits}}/4$, $m = 8$, single seed. The data reveal *two* regimes of LLL failure: (1) secp256k1 specifically at 256 bits, while other 256-bit curves succeed; (2) all ≥ 384-bit curves *uniformly* fail, suggesting an iteration-cap issue rather than a curve-specific property.

4.1 Two distinct degeneracies

Failure mode A (curve-specific): secp256k1 at 256 bits. P-256 and brainpoolP256r1 — both 256-bit curves with similar basis-entry sizes — succeed at LLL while secp256k1 fails. This is a curve-specific arithmetic phenomenon, not bit-length-driven.

Failure mode B (uniform): all ≥ 384-bit curves. P-384, brainpoolP384r1, and P-521 all fail uniformly. None has the special arithmetic shape of secp256k1's near-power-of-2 n; P-384 in particular is the NIST "Solinas" prime with no Koblitz structure. The uniformity strongly suggests an *implementation-side* cause (LLL iteration cap insufficient at large bit-lengths) rather than a curve-side cause.

5 Reframed open questions

The data narrow the open questions to two distinct ones, one per failure mode:

5.1 Question A: why does secp256k1 alone fail at 256 bits?

P-256 and brainpoolP256r1 share secp256k1's bit-length. None of the three is supersingular; all are ordinary with prime group order. The discriminating features:

- secp256k1 has $j = 0$ (CM by $\mathbb{Z}[\omega]$); P-256 and brainpoolP256r1 are non-CM (generic j).
- secp256k1's coefficient $a = 0$; P-256's $a = -3$; brainpoolP256r1's a is a specific 256-bit value.
- secp256k1's group order has a near-power-of-2 form (refuted as the sole cause by the smaller Koblitz curves, but it may still contribute at 256 bits specifically).

The most natural follow-up: probe additional $j = 0$ curves at 256 bits to see whether the failure tracks $j = 0$. Candidates: custom-constructed $j = 0$ Koblitz curves over different near-256-bit primes.

5.2 Question B: why does LLL uniformly fail at ≥ 384 bits?

The LLL iteration cap in our implementation is $\text{cap} = 500 \cdot d^2 \cdot 8 + 10^4$ where d is the lattice dimension. For $d = 10$ $(= m + 2 = 8 + 2)$, this gives $\text{cap} \approx 4.1 \times 10^5$.

Classical LLL convergence is $O(d^4 \log B)$ where B is the max entry norm. With $d = 10$ and $B \approx n^2 \approx 2^{2 \cdot 384} \approx 2^{768}$ for P-384, the expected iteration count is $\sim 10000 \cdot 768 \approx 7.7 \times 10^6$ — **a factor of 20 above the cap**. This strongly suggests the cap is simply too small for ≥ 384-bit curves.

The cleanest single test: raise the cap to 10^7 and re-run on P-384. If LLL converges, the cap was the cause. Then the cap formula should be $O(d^2 \log B)$ rather than $O(d^2)$.

5.3 Why secp256k1 alone fails at 256 bits (but not P-256)

P-256's n and secp256k1's n are both 256-bit. Their basis B-matrices have entries of comparable bit-length. So why does LLL converge in ∼700 ms for P-256 but hit the cap for secp256k1?

We do not have a closed-form answer. The candidate sources are: the specific (a_i, t_i) distribution induced by secp256k1's trace $t \approx 2^{129}$; the $\mathbb{Z}[\omega]$-decomposition of Frobenius $\pi = a + b\omega$; the $r_i = (kG)_x \bmod n$ distribution being narrower for secp256k1 (where n is very close to p) than for P-256.

A theoretical Gram–Schmidt analysis is sketched in the companion note [7] but does not yet pin down the mechanism precisely. The next concrete step is to compute the Gram-matrix eigenvalue spectrum empirically for both curves at the same parameter set.

6 Implications and outlook

The reframed open question is sharper than the original. The phenomenon's cause is somewhere in the intersection of: specific arithmetic of secp256k1's (p, n, t, a, b) tuple, the specific cap parameter in the LLL implementation, and the basis- construction step's sensitivity to coefficient size.

Importantly, the empirical observation *remains valid* as a practical attack-cost benchmark: any HNP attack reporting LLL costs on a single curve cannot extrapolate to other curves without revalidation. The refuted hypothesis only ruled out one *theoretical* explanation; it did not weaken the *empirical* claim.

7 Implications

7.1 For practical ECDSA attacks

The observed degeneracy is *not* a security claim about secp256k1: an attacker with biased ECDSA signatures and willing to use BKZ-β (with β chosen large enough) will still recover the key. LLL is the *first-line* attack; the degeneracy delays it but does not block more thorough lattice reduction. The observed cost difference — a factor of $\sim 15\times$ between LLL convergence on P-256 and LLL failure on secp256k1 — does not translate into a security margin.

7.2 For HNP-attack benchmarking literature

We note that published HNP-on-ECDSA papers (e.g., [2, 3, 4]) often demonstrate the attack on a single curve (often P-256 or secp256k1) without head-to-head curve comparison. Our observation suggests that LLL-convergence benchmarks are *curve-specific* in a way that has not been previously documented; reported attack costs that generalise across curves should be treated with caution.

7.3 For the GLV-aware HNP programme

This phenomenon affects the proposed GLV-aware Hidden-Number Problem variant for secp256k1 [7, §4], which plans to use a lattice of similar structure (with the addition of half-scalar bookkeeping for the GLV decomposition). If the underlying near-power-of-2 mechanism propagates, the GLV-aware secp256k1 attack will need a different reduction strategy.

8 Open questions

1. **Theoretical**: derive the precise Gram–Schmidt eigenvalue structure of the Boneh–Venkatesan basis for n of near-power-of-2 form. Quantify the spectral gap (or its absence) that LLL's size-reduction inequality measures.

2. **Empirical**: replicate the head-to-head probe across all NIST curves (P-192, P-224, P-256, P-384, P-521), the brainpool family, and the Koblitz family (secp192k1, secp224k1, secp256k1). Tabulate convergence behaviour as a function of how close n is to a power of 2.

3. **Algorithmic**: identify the smallest BKZ block size β that converges where LLL fails. Cost growth from LLL to BKZ-β as a function of β.

4. **Curve-design**: are there ECC curve standardisation criteria that should explicitly avoid near-power-of-2 group orders for HNP-robustness reasons? Currently no such criterion appears in published curve-selection rationales.

9 Conclusion

We have documented a curve-specific LLL-convergence asymmetry between secp256k1 and P-256 at standard HNP-attack parameters. The phenomenon is empirically robust (3-of-3 vs 0-of-3 across three independent seeds) and reproducible via the companion code. It does not affect ECDSA security in practice (BKZ recovers where LLL fails) but it does suggest that HNP benchmark costs reported on one curve do not automatically transfer to another. The underlying mechanism — whether near-power-of-2 n is indeed the cause, or whether some deeper basis-structure feature is at play — is an open question for further work.

References

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